



Lab Task 3

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	
Clarity						
Content Quality						
Spelling & Grammar						
Organization and Formatting						
Total obtained mark						
Comments						

Semester: Spring / Fall

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Batch: 46

Section:B

Course Code: CSE 124

Course Name: Object Oriented Programming

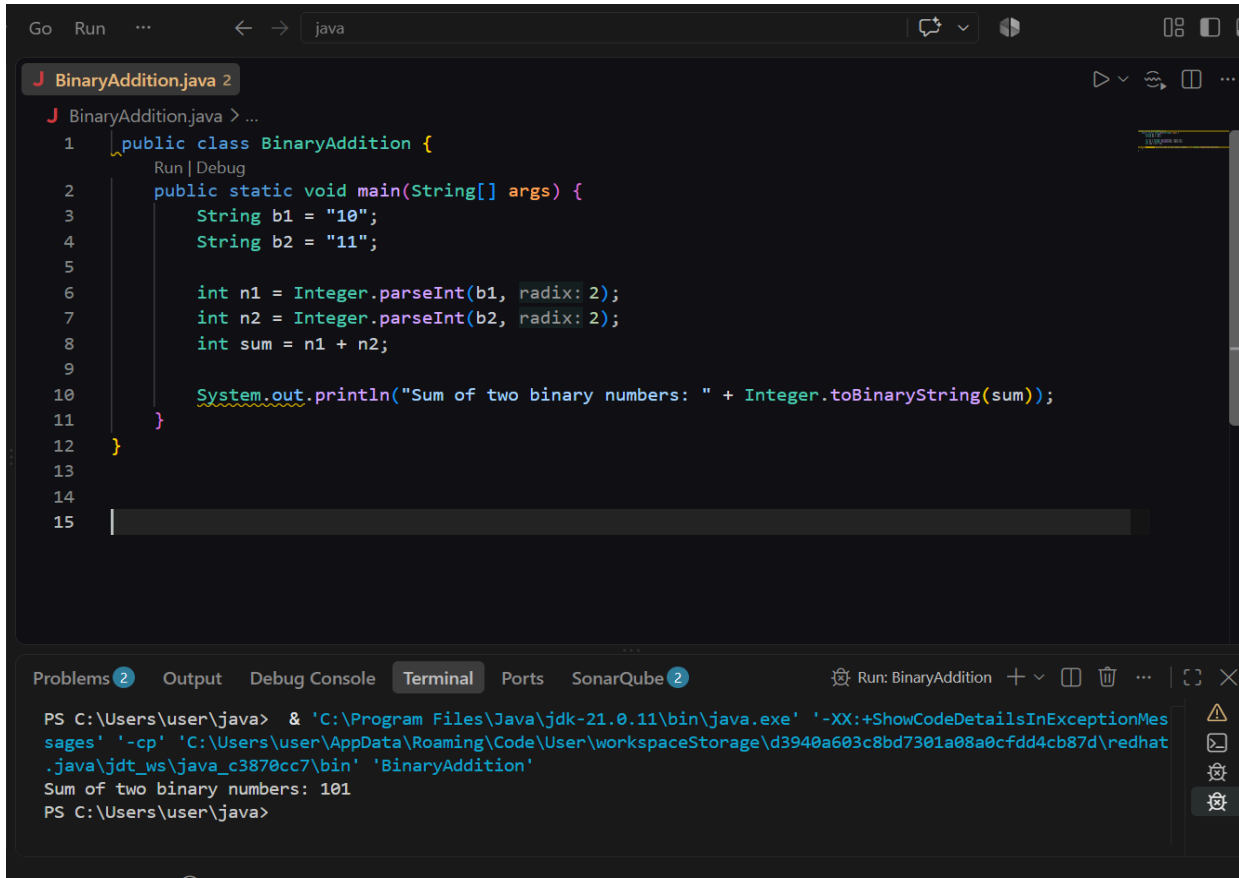
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Designation:Lecturer

Submission Date: 15 / 08 /2026

Experiment No : 17

Experiment Name : Binary Addition .



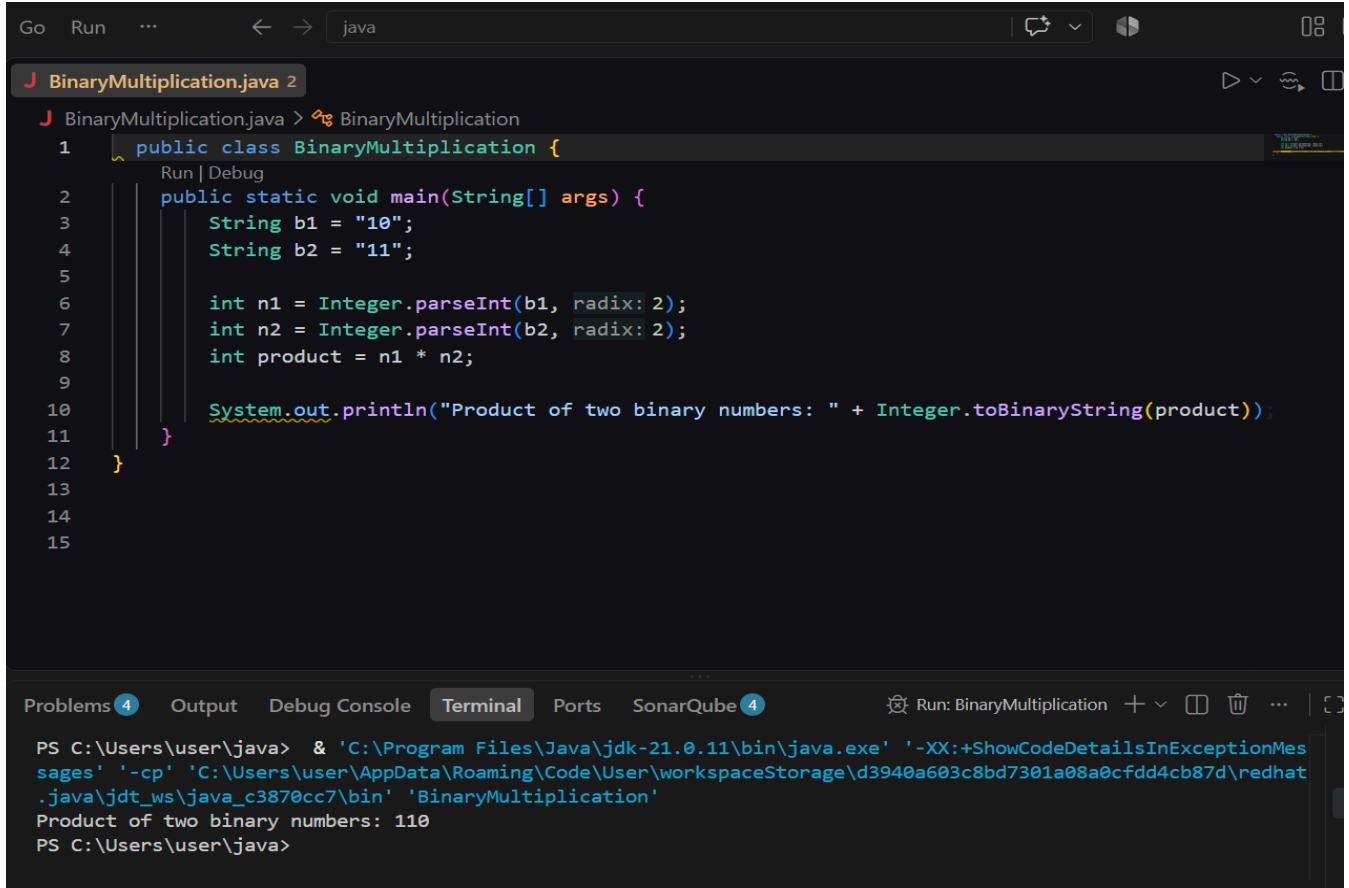
```
Go Run ... < -> java
J BinaryAddition.java 2
J BinaryAddition.java > ...
1 public class BinaryAddition {
   Run | Debug
2     public static void main(String[] args) {
3         String b1 = "10";
4         String b2 = "11";
5
6         int n1 = Integer.parseInt(b1, radix: 2);
7         int n2 = Integer.parseInt(b2, radix: 2);
8         int sum = n1 + n2;
9
10        System.out.println("Sum of two binary numbers: " + Integer.toBinaryString(sum));
11    }
12 }
13
14
15

Problems 2 Output Debug Console Terminal Ports SonarQube 2
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'BinaryAddition'
Sum of two binary numbers: 101
PS C:\Users\user\java>
```

Discussion: I first converted both binary numbers into decimal. Then I added them normally and converted the result back to binary. This way is simple and works fine.

Experiment No:18

Experiment Name : Binary Multiplication .



The screenshot displays an IDE window with a Java file named `BinaryMultiplication.java`. The code defines a public class `BinaryMultiplication` with a `main` method. Inside `main`, two binary strings `"10"` and `"11"` are assigned to `b1` and `b2` respectively. These are converted to integers `n1` and `n2` using `Integer.parseInt` with a radix of 2. The product is calculated as `n1 * n2` and then converted back to a binary string using `Integer.toBinaryString`. The result is printed to the console.

```
1 public class BinaryMultiplication {  
2     public static void main(String[] args) {  
3         String b1 = "10";  
4         String b2 = "11";  
5  
6         int n1 = Integer.parseInt(b1, radix: 2);  
7         int n2 = Integer.parseInt(b2, radix: 2);  
8         int product = n1 * n2;  
9  
10        System.out.println("Product of two binary numbers: " + Integer.toBinaryString(product));  
11    }  
12 }  
13  
14  
15
```

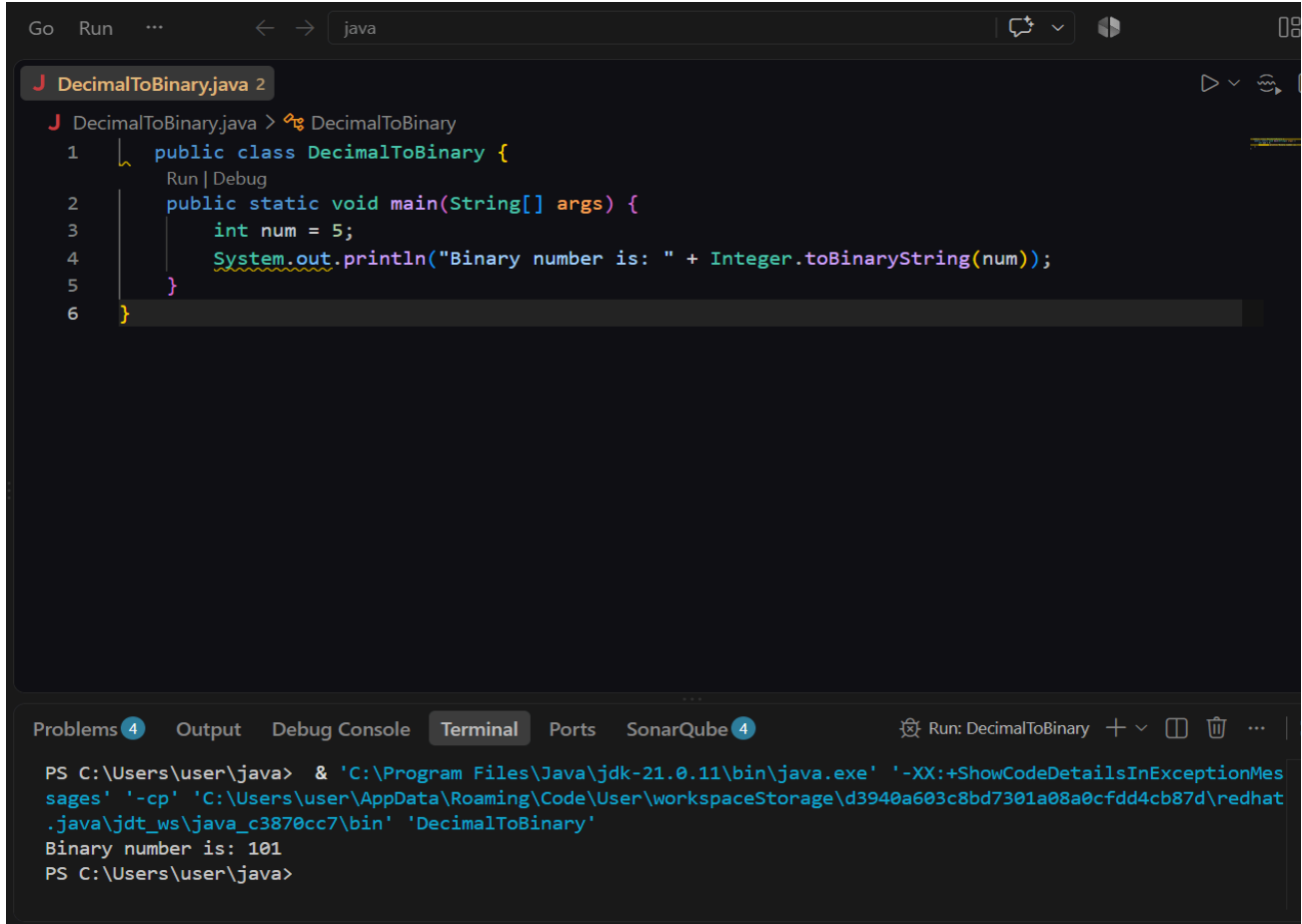
The terminal output at the bottom shows the command to run the program and the resulting output:

```
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'BinaryMultiplication'  
Product of two binary numbers: 110  
PS C:\Users\user\java>
```

Discussion : I converted the two binary numbers to decimal, multiplied them, and then changed the product back to binary, Easy method.

Experiment No : 19

Experiment Name : Decimal To Binary Convert .

A screenshot of an IDE window titled 'DecimalToBinary.java 2'. The code defines a public class 'DecimalToBinary' with a static 'main' method. Inside 'main', an integer 'num' is set to 5, and a line of code uses 'Integer.toBinaryString(num)' to convert it to a binary string, which is then printed. The IDE interface includes a top toolbar with 'Go', 'Run', and other icons, and a bottom panel with tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', 'Ports', and 'SonarQube'. The 'Terminal' tab is active, showing the command prompt output of the program execution.

```
Go Run ... < > java
J DecimalToBinary.java 2
J DecimalToBinary.java > DecimalToBinary
1 public class DecimalToBinary {
2     Run | Debug
3     public static void main(String[] args) {
4         int num = 5;
5         System.out.println("Binary number is: " + Integer.toBinaryString(num));
6     }
}
```

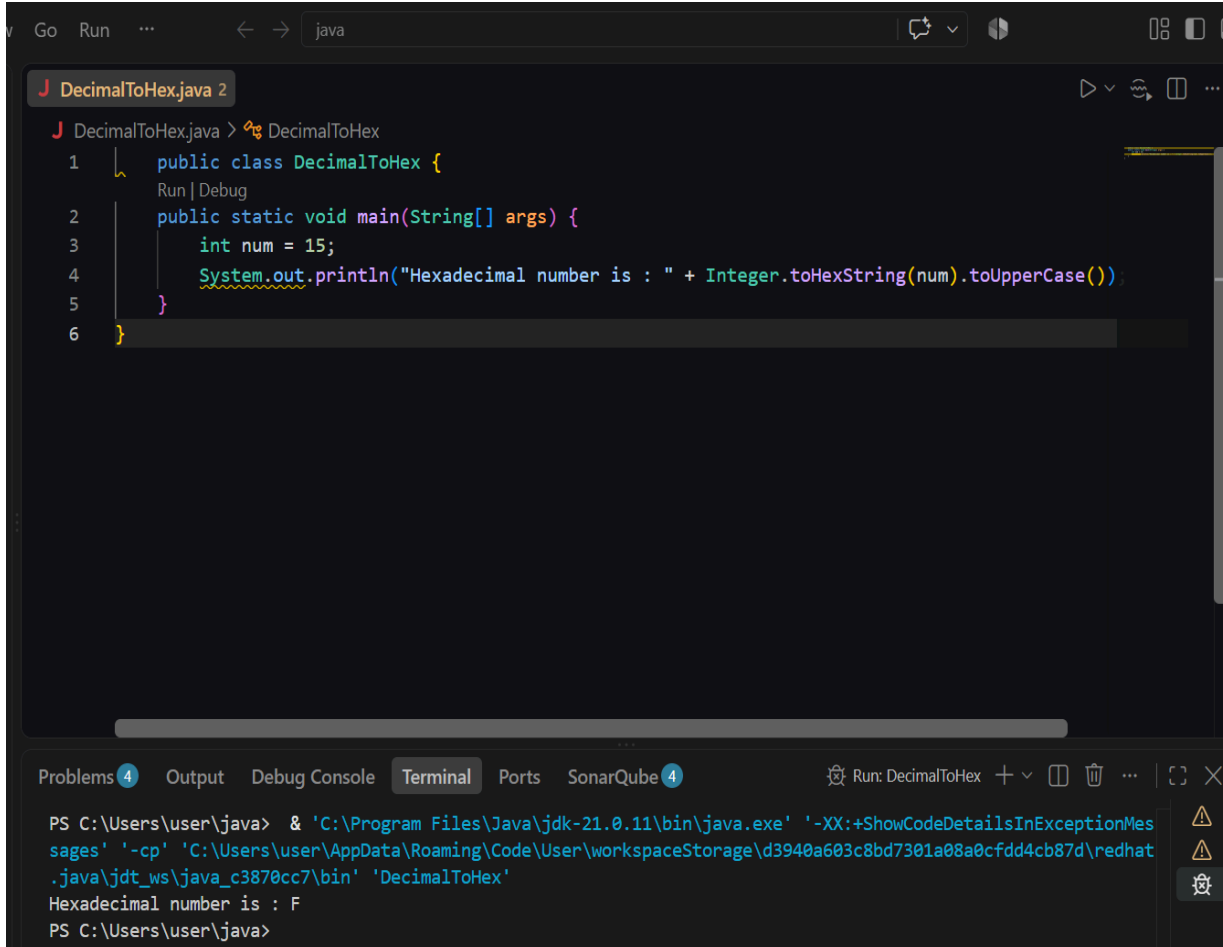
Problems 4 Output Debug Console Terminal Ports SonarQube 4 Run: DecimalToBinary + - [] [] ...

```
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'DecimalToBinary'
Binary number is: 101
PS C:\Users\user\java>
```

Discussion: Java has a ready method called `toBinaryString()`. I just used that to convert the decimal number to binary, Nothing extra needed.

Experiment No : 20

Experiment Name : Decimal To Hexadecimal .



The screenshot shows an IDE with a Java file named `DecimalToHex.java`. The code defines a class `DecimalToHex` with a `main` method. Inside `main`, a variable `num` is set to 15, and a message is printed using `Integer.toHexString(num).toUpperCase()`. Below the code editor, the `Terminal` tab is active, showing the command to run the program and the output: "Hexadecimal number is : F".

```
DecimalToHex.java 2
J DecimalToHex.java > DecimalToHex
1 public class DecimalToHex {
  Run | Debug
2 public static void main(String[] args) {
3     int num = 15;
4     System.out.println("Hexadecimal number is : " + Integer.toHexString(num).toUpperCase());
5 }
6 }
```

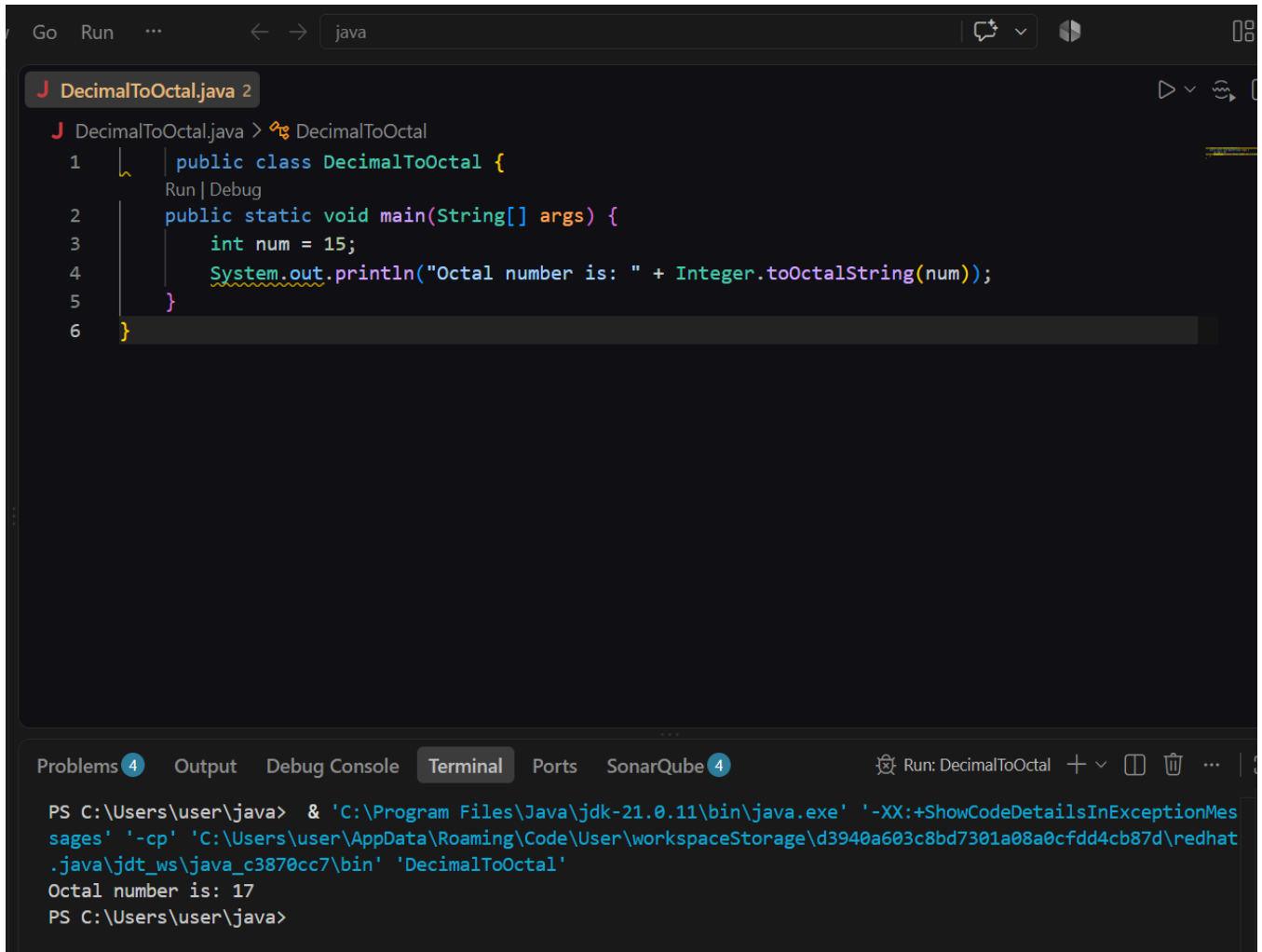
Problems 4 Output Debug Console Terminal Ports SonarQube 4 Run: DecimalToHex

```
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'DecimalToHex'
Hexadecimal number is : F
PS C:\Users\user\java>
```

Discussion: I used `toHexString()` to convert decimal to hexadecimal. I also made it uppercase so the output matches the expected result.

Experiment No : 21

Experiment Name : Decimal To Octal Converter .

A screenshot of an IDE window titled 'DecimalToOctal.java 2'. The code defines a class 'DecimalToOctal' with a 'main' method. Inside 'main', a variable 'num' is set to 15, and 'System.out.println' is used to print the octal string of 'num' using 'Integer.toOctalString(num)'. Below the code editor, the 'Terminal' tab is active, showing the command prompt execution of the program, which outputs 'Octal number is: 17'.

```
DecimalToOctal.java 2
DecimalToOctal.java > DecimalToOctal
1  public class DecimalToOctal {
   Run | Debug
2  public static void main(String[] args) {
3      int num = 15;
4      System.out.println("Octal number is: " + Integer.toOctalString(num));
5  }
6  }
```

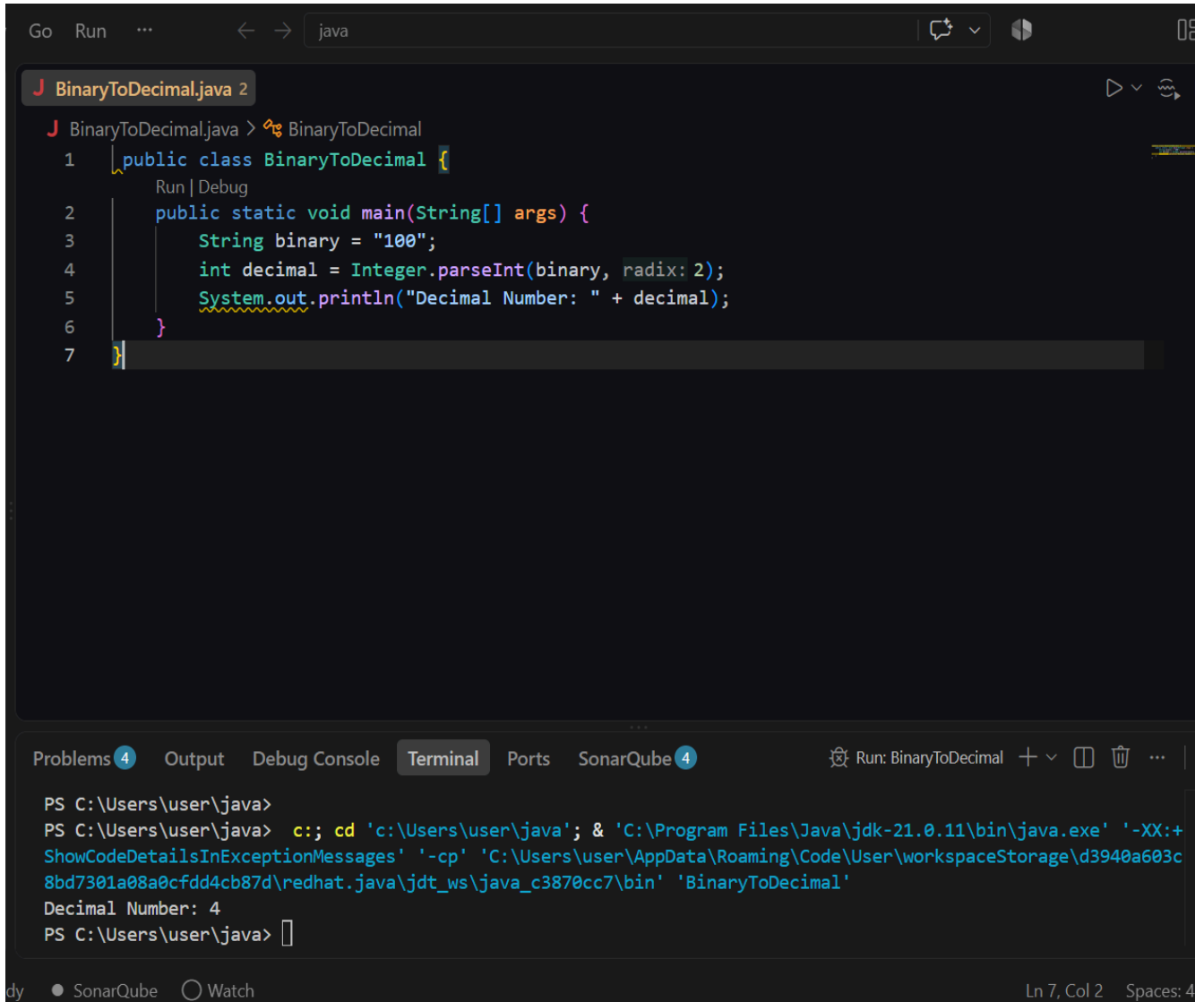
Problems 4 Output Debug Console Terminal Ports SonarQube 4 Run: DecimalToOctal

```
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'DecimalToOctal'
Octal number is: 17
PS C:\Users\user\java>
```

Discussion: I used the `to OctalString()` method to convert the decimal number into octal ,Very straightforward.

Experiment No : 22

Experiment Name : Binary To Decimal Converter .

The screenshot shows an IDE with a dark theme. The top toolbar includes 'Go', 'Run', and a search icon. The file explorer on the left shows 'BinaryToDecimal.java 2'. The editor displays the following Java code:

```
1 public class BinaryToDecimal {  
2     public static void main(String[] args) {  
3         String binary = "100";  
4         int decimal = Integer.parseInt(binary, radix: 2);  
5         System.out.println("Decimal Number: " + decimal);  
6     }  
7 }
```

The bottom panel has tabs for 'Problems' (4), 'Output', 'Debug Console', 'Terminal' (active), 'Ports', and 'SonarQube' (4). The terminal shows the command to run the program and its output:

```
PS C:\Users\user\java>  
PS C:\Users\user\java> c:; cd 'c:\Users\user\java'; & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+  
ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c  
8bd7301a08a0c added4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'BinaryToDecimal'  
Decimal Number: 4  
PS C:\Users\user\java>
```

The status bar at the bottom indicates 'Ln 7, Col 2' and 'Spaces: 4'.

Discussion : I used `parseInt` with base 2 to convert the binary string into decimal. It works directly.

Experiment No: 23

Experiment Name : Binary To Hexadecimal Converter .

The screenshot shows an IDE window titled "BinaryToHex.java 2". The code defines a class "BinaryToHex" with a main method. Inside the main method, a string "1101" is assigned to a variable named "binary". This string is then converted to a decimal integer using "Integer.parseInt(binary, radix: 2)". Finally, the decimal value is converted back to a hexadecimal string using "Integer.toHexString(decimal).toUpperCase()" and printed to the console. Below the code editor, there are tabs for "Problems", "Output", "Debug Console", "Terminal", "Ports", and "SonarQube". The "Terminal" tab is active, displaying the command prompt output where the program was executed successfully, resulting in the hexadecimal value "D".

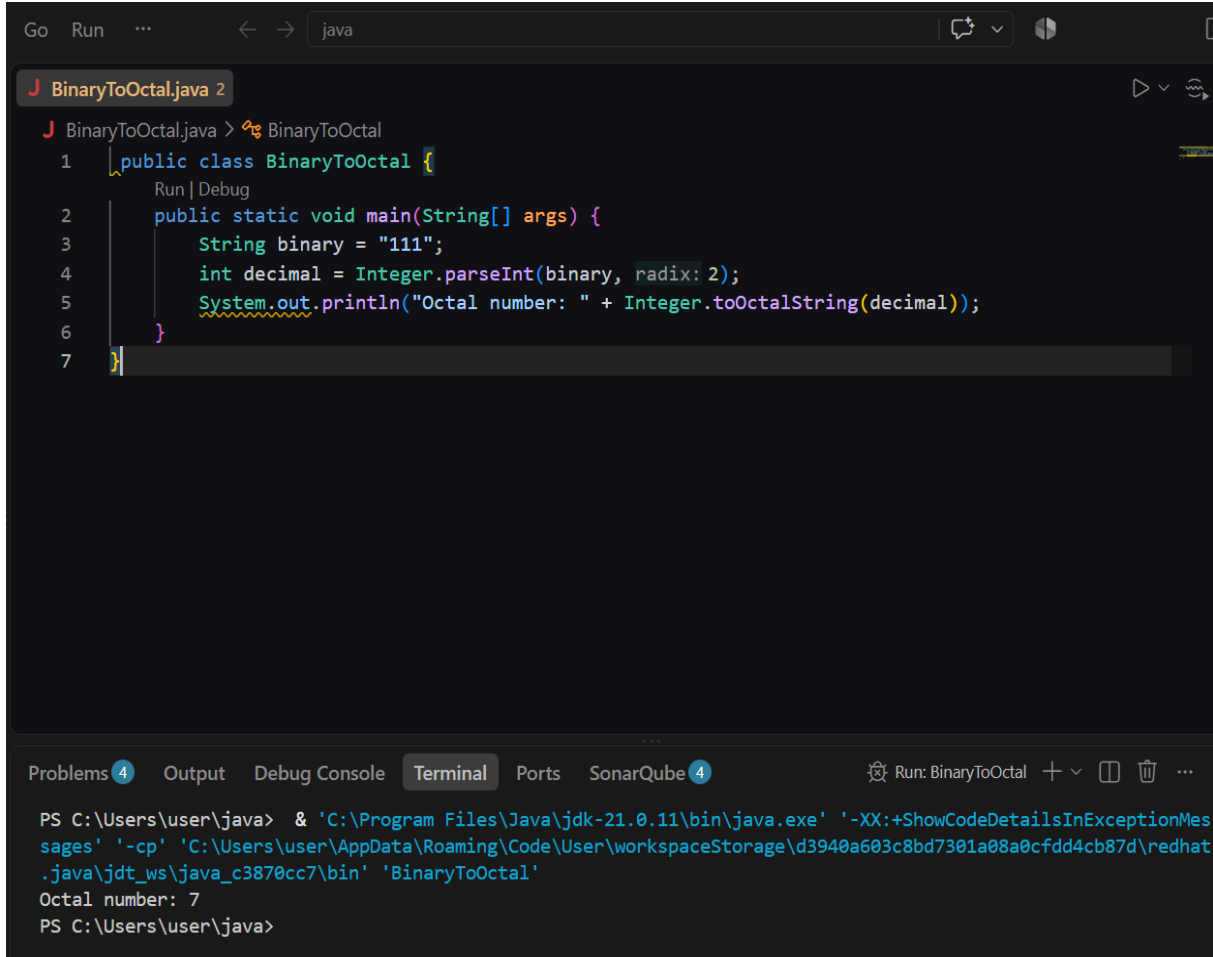
```
1 public class BinaryToHex {  
2     public static void main(String[] args) {  
3         String binary = "1101";  
4         int decimal = Integer.parseInt(binary, radix: 2);  
5         System.out.println("Hexadecimal value: " + Integer.toHexString(decimal).toUpperCase());  
6     }  
7 }
```

PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'BinaryToHex'
Hexadecimal value: D
PS C:\Users\user\java>

Discussion: First I converted binary to decimal, then converted that decimal to hexadecimal. Did it in two steps.

Experiment No : 24

Experiment Name : Binary To Octal Converter .



```
Go Run ... < > java
J BinaryToOctal.java 2
J BinaryToOctal.java > BinaryToOctal
1 public class BinaryToOctal {
    Run | Debug
2     public static void main(String[] args) {
3         String binary = "111";
4         int decimal = Integer.parseInt(binary, radix: 2);
5         System.out.println("Octal number: " + Integer.toOctalString(decimal));
6     }
7 }
```

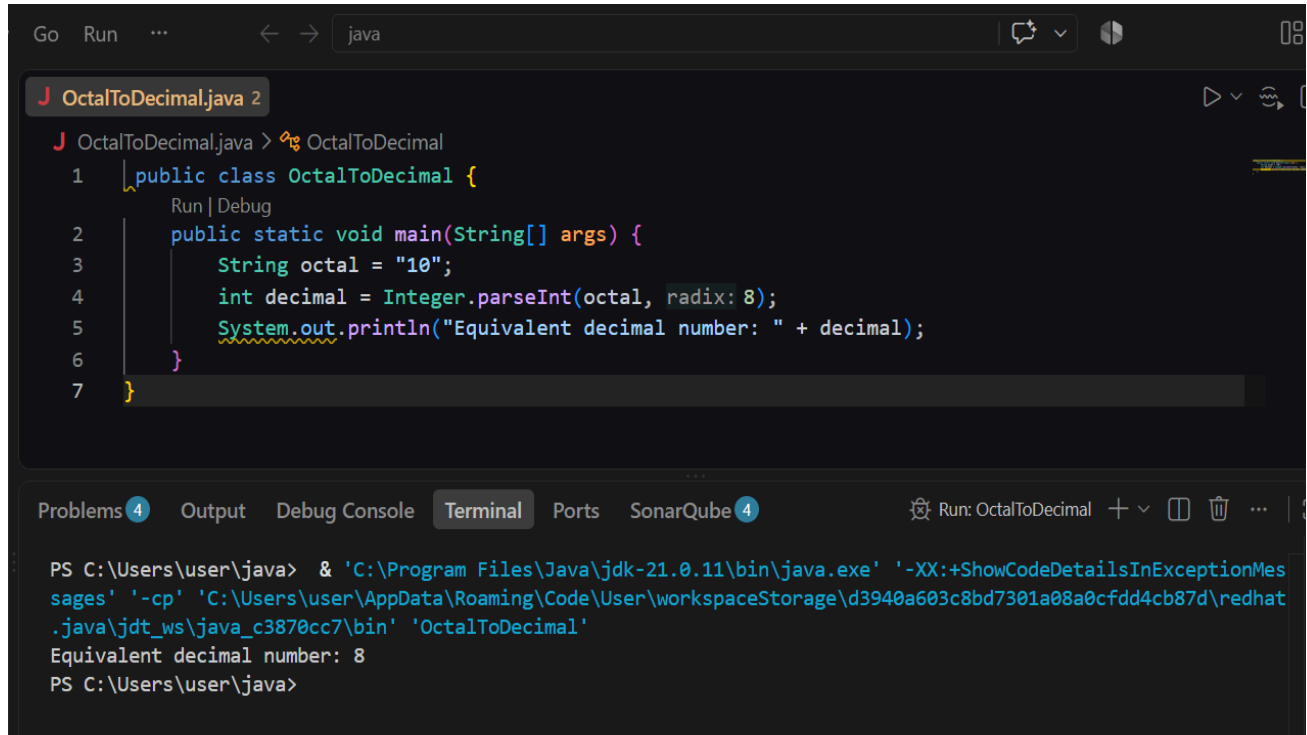
Problems 4 Output Debug Console Terminal Ports SonarQube 4 Run: BinaryToOctal + - [] [] ...

```
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'BinaryToOctal'
Octal number: 7
PS C:\Users\user\java>
```

Discussion: I changed binary to decimal first and then converted the decimal value to octal. Simple way.

Experiment No : 25

Experiment Name : Octal To Decimal Converter .



The screenshot displays an IDE window with a Java file named `OctalToDecimal.java`. The code defines a class `OctalToDecimal` with a `main` method that takes an array of strings `args`. Inside the `main` method, a string `octal` is assigned the value `"10"`, an integer `decimal` is calculated using `Integer.parseInt(octal, radix: 8)`, and the result is printed to the console using `System.out.println("Equivalent decimal number: " + decimal)`.

Below the code editor, the `Terminal` tab is active, showing the command prompt output. The command executed is `PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'OctalToDecimal'`. The output shows `Equivalent decimal number: 8`.

```
Go Run ... < -> java
J OctalToDecimal.java 2
J OctalToDecimal.java > OctalToDecimal
1 public class OctalToDecimal {
  Run | Debug
2   public static void main(String[] args) {
3     String octal = "10";
4     int decimal = Integer.parseInt(octal, radix: 8);
5     System.out.println("Equivalent decimal number: " + decimal);
6   }
7 }

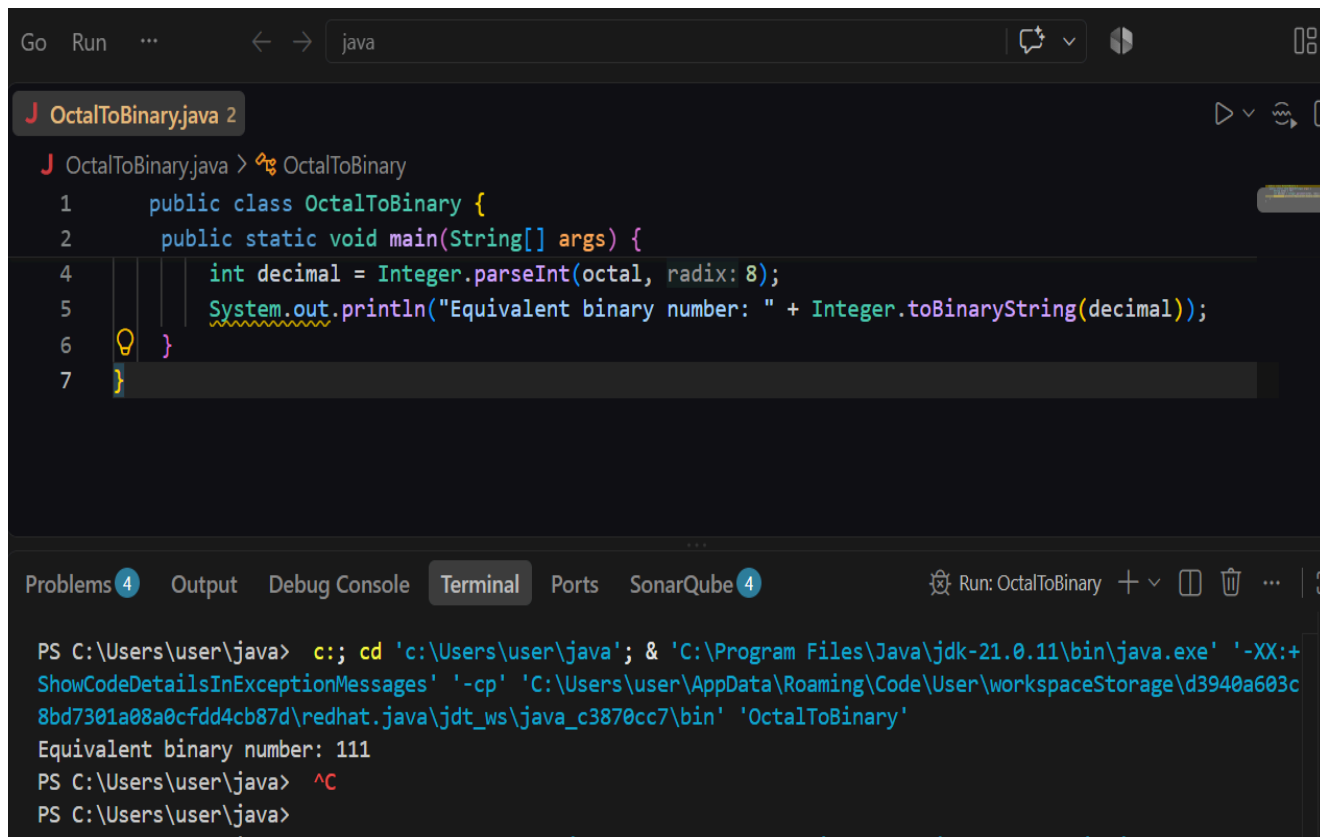
Problems 4 Output Debug Console Terminal Ports SonarQube 4
Run: OctalToDecimal + - [ ] [ ] ...

PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMes
sages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat
.java\jdt_ws\java_c3870cc7\bin' 'OctalToDecimal'
Equivalent decimal number: 8
PS C:\Users\user\java>
```

Discussion: I used `parseInt` with base 8 to convert the octal number into decimal. It works fine.

Experiment No : 26

Experiment Name : Octal To Binary Converter .



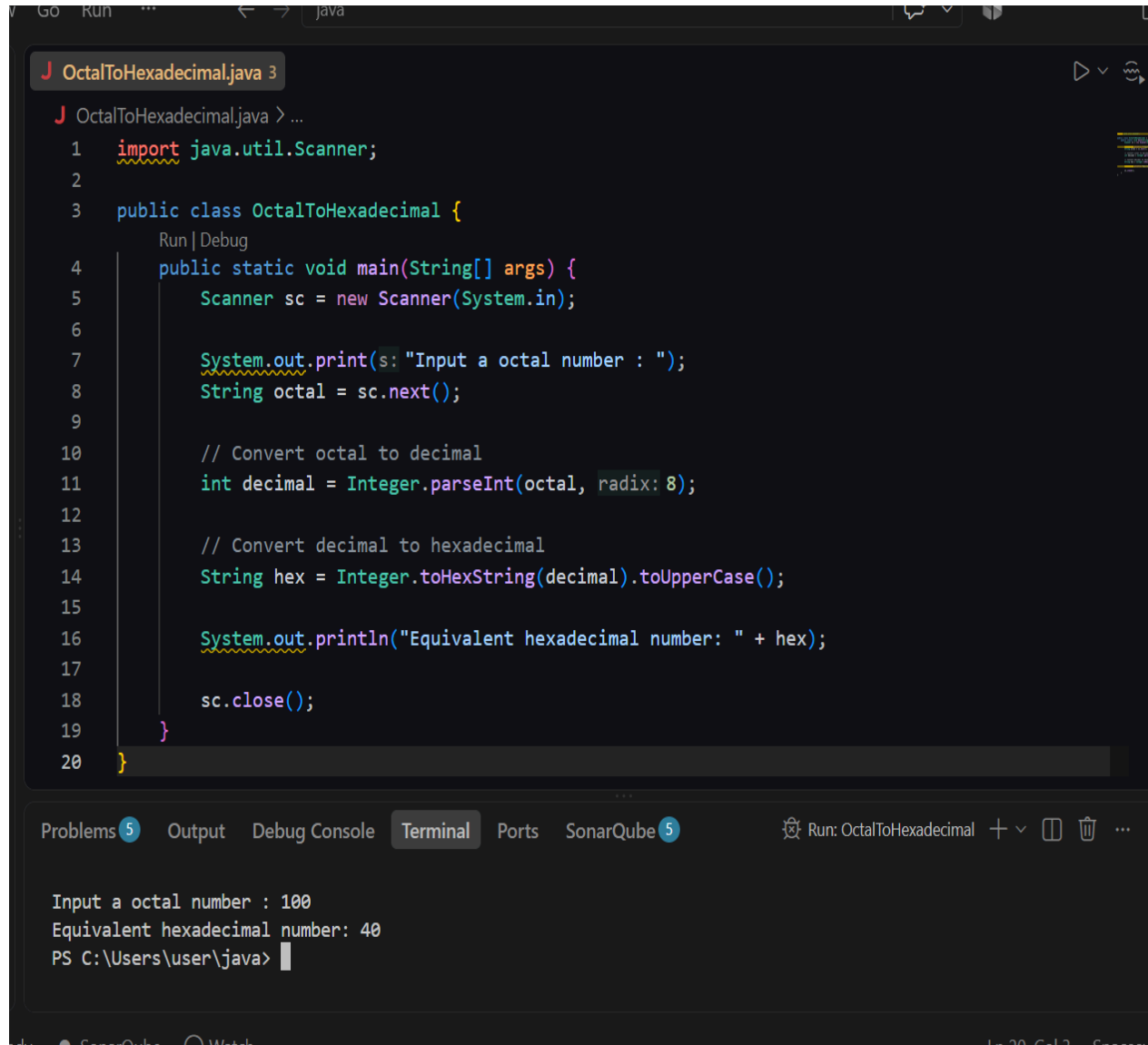
```
Go Run ... java
J OctalToBinary.java 2
J OctalToBinary.java > OctalToBinary
1 public class OctalToBinary {
2     public static void main(String[] args) {
4         int decimal = Integer.parseInt(octal, radix: 8);
5         System.out.println("Equivalent binary number: " + Integer.toBinaryString(decimal));
6     }
7 }
```

```
Problems 4 Output Debug Console Terminal Ports SonarQube 4
Run: OctalToBinary
PS C:\Users\user\java> c::; cd 'c:\Users\user\java'; & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+
ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c
8bd7301a08a0cfd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'OctalToBinary'
Equivalent binary number: 111
PS C:\Users\user\java> ^C
PS C:\Users\user\java>
```

Discussion: I first converted octal to decimal and then changed the decimal to binary. This two-step method is easy.

Experiment No : 27

Experiment Name : Octal To Hexadecimal Converter .



```
OctalToHexadecimal.java 3
OctalToHexadecimal.java > ...
1  import java.util.Scanner;
2
3  public class OctalToHexadecimal {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6
7          System.out.print(s: "Input a octal number : ");
8          String octal = sc.next();
9
10         // Convert octal to decimal
11         int decimal = Integer.parseInt(octal, radix: 8);
12
13         // Convert decimal to hexadecimal
14         String hex = Integer.toHexString(decimal).toUpperCase();
15
16         System.out.println("Equivalent hexadecimal number: " + hex);
17
18         sc.close();
19     }
20 }
```

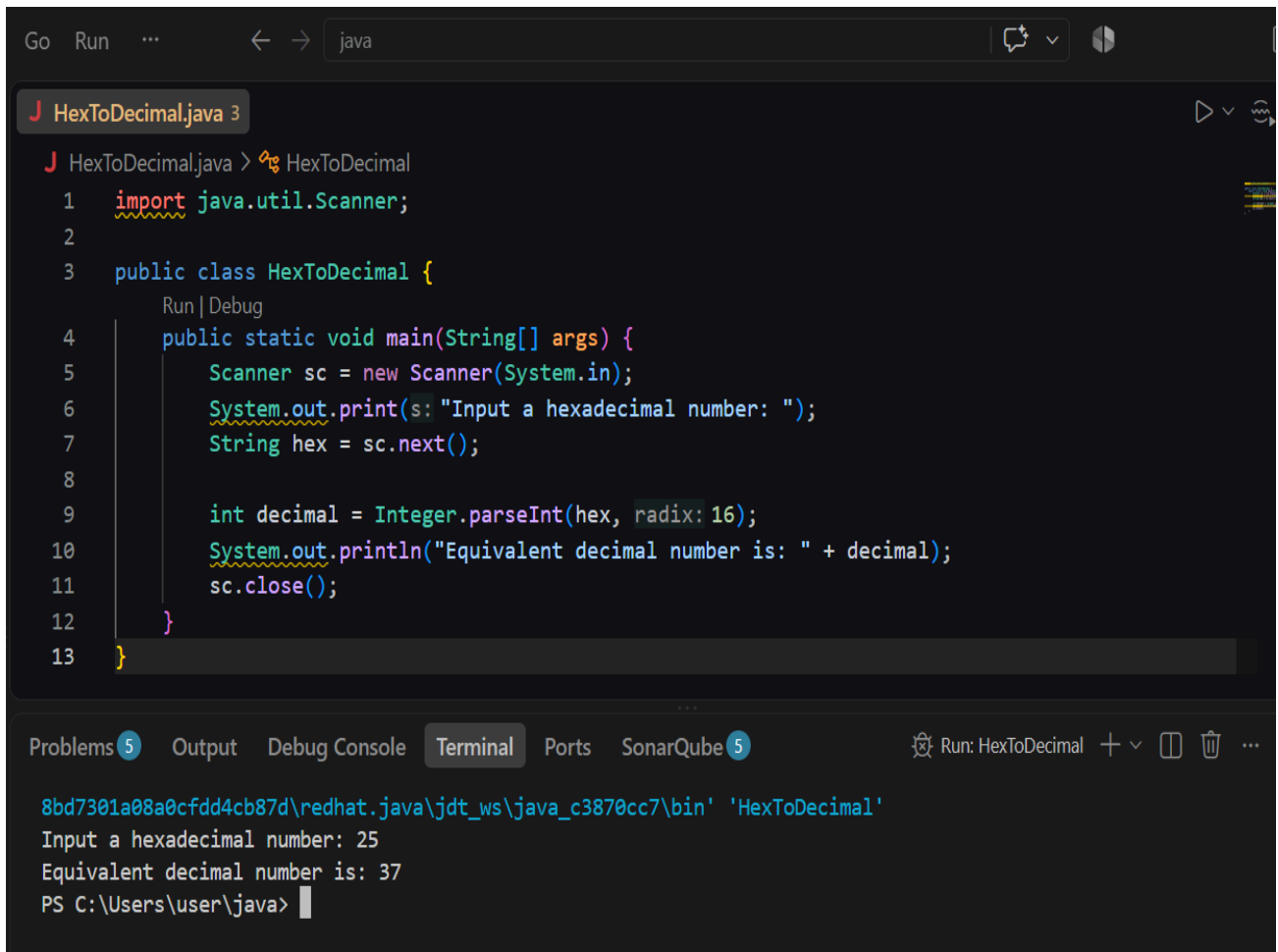
Problems 5 Output Debug Console Terminal Ports SonarQube 5 Run: OctalToHexadecimal + - [] [] ...

```
Input a octal number : 100
Equivalent hexadecimal number: 40
PS C:\Users\user\java>
```

Discussion: Converted octal to decimal first, then decimal to hexadecimal. Made it uppercase for the output.

Experiment No : 28

Experiment Name : Hexadecimal To Decimal Converter .



The screenshot shows an IDE with a Java file named `HexToDecimal.java`. The code is as follows:

```
1 import java.util.Scanner;
2
3 public class HexToDecimal {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         System.out.print(s: "Input a hexadecimal number: ");
7         String hex = sc.next();
8
9         int decimal = Integer.parseInt(hex, radix: 16);
10        System.out.println("Equivalent decimal number is: " + decimal);
11        sc.close();
12    }
13 }
```

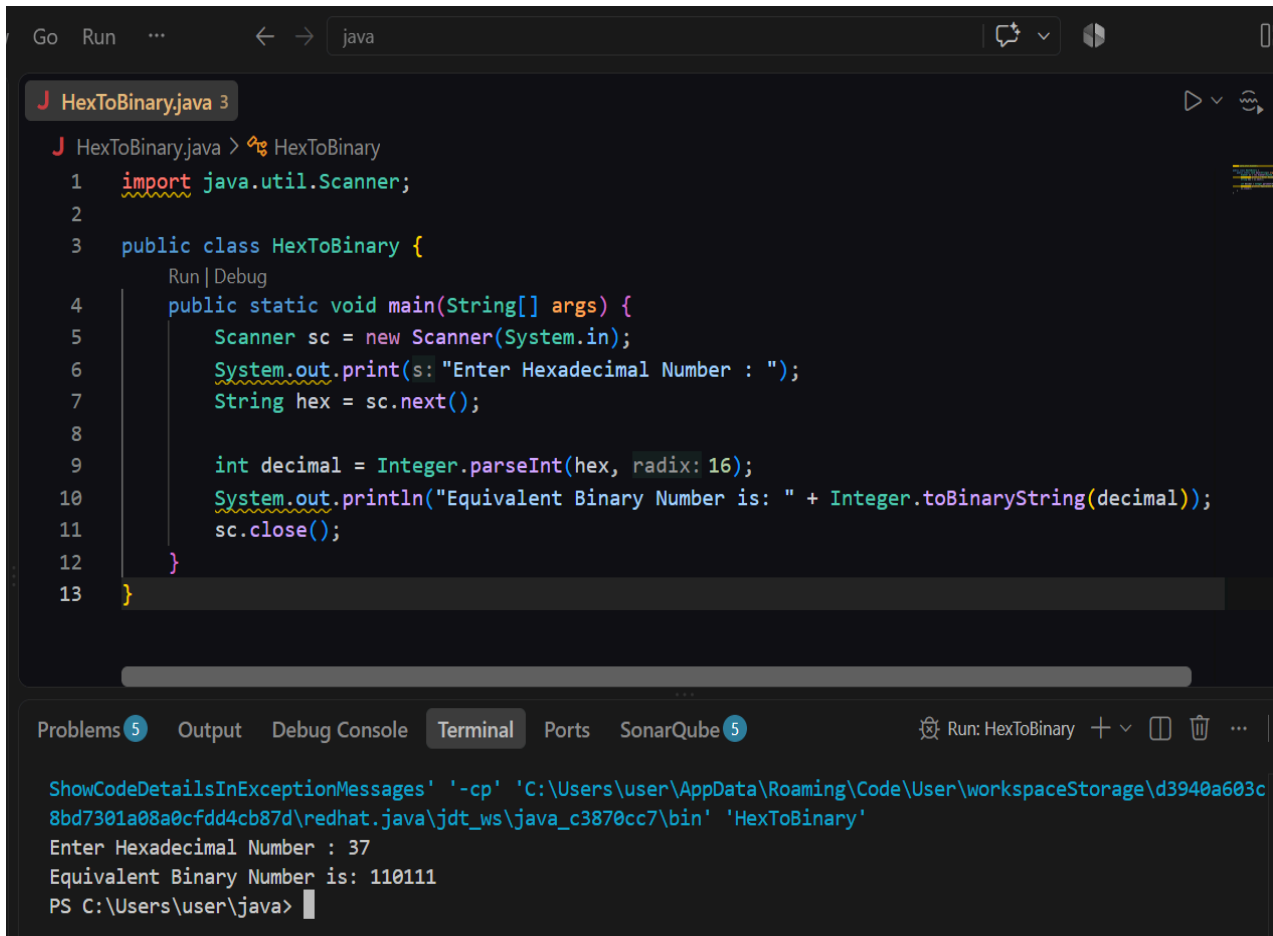
The IDE's bottom panel shows the `Terminal` tab with the following output:

```
8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'HexToDecimal'
Input a hexadecimal number: 25
Equivalent decimal number is: 37
PS C:\Users\user\java>
```

Discussion: I used `parseInt` with base 16 to convert the hexadecimal number directly into decimal.

Experiment No : 29

Experiment Name : Hexadecimal To Binary Converter .



The screenshot shows an IDE window with a Java file named `HexToBinary.java`. The code is as follows:

```
1  import java.util.Scanner;
2
3  public class HexToBinary {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print(s: "Enter Hexadecimal Number : ");
7          String hex = sc.next();
8
9          int decimal = Integer.parseInt(hex, radix: 16);
10         System.out.println("Equivalent Binary Number is: " + Integer.toBinaryString(decimal));
11         sc.close();
12     }
13 }
```

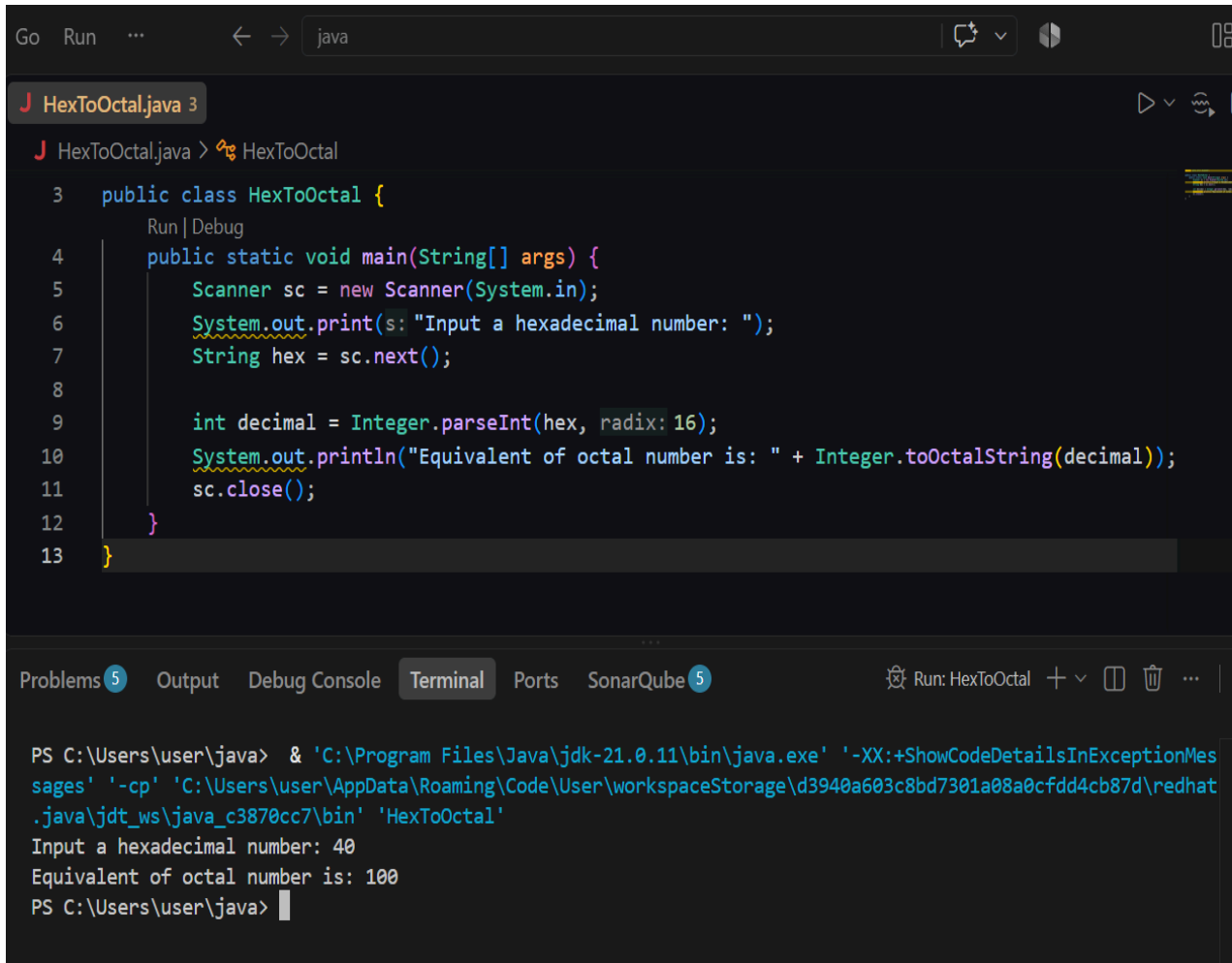
Below the code editor, the `Terminal` tab is active, showing the execution of the program:

```
ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'HexToBinary'
Enter Hexadecimal Number : 37
Equivalent Binary Number is: 110111
PS C:\Users\user\java>
```

Discussion: First converted hexadecimal to decimal, then converted that decimal value to binary. Did it in two steps.

Experiment No: 30

Experiment Name : Hexadecimal To Octal Converter .



The screenshot displays an IDE with a Java file named `HexToOctal.java`. The code defines a `HexToOctal` class with a `main` method. The `main` method uses a `Scanner` to take a hexadecimal input, converts it to a decimal integer using `Integer.parseInt` with a radix of 16, and then converts the decimal to an octal string using `Integer.toOctalString`. The output shows the conversion of the hexadecimal number 40 to the octal number 100.

```
3 public class HexToOctal {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         System.out.print(s: "Input a hexadecimal number: ");
7         String hex = sc.next();
8
9         int decimal = Integer.parseInt(hex, Radix: 16);
10        System.out.println("Equivalent of octal number is: " + Integer.toOctalString(decimal));
11        sc.close();
12    }
13 }
```

Terminal Output:

```
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'HexToOctal'
Input a hexadecimal number: 40
Equivalent of octal number is: 100
PS C:\Users\user\java>
```

Discussion: I converted hexadecimal to decimal first and then changed the decimal to octal. Simple and clear.